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CEF440: INTERNET PROGRAMMING AND MOBILE PROGRAMMING

PROJECT TITLE: DESIGN AND IMPLEMENTATION OF A MOBILE APP FOR COLLECTION OF USER EXPERIENCE DATA FROM MOBILE NETWORK SUBSCRIBERS.

TASK 2: REQUIREMENT GATHERING

GROUP 2

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I. INTRODUCTION

Requirement gathering is a critical phase in system development, as it defines what the system must achieve and how it should function. For this project, which focuses on the design and implementation of a mobile application for collecting user experience data from mobile network subscribers, requirement gathering ensures that both user needs and system functionalities are clearly understood before development begins.

This phase involved identifying stakeholders, selecting appropriate data collection techniques, gathering and cleaning data, and assessing potential user reluctance toward using the proposed system.

II. STAKEHOLDER IDENTIFICATION

Stakeholders are individuals or organizations that have an interest in the system or are affected by its implementation. We have 2 types of stake holders.

1) Primary Stakeholders

Mobile Network Subscribers (Users):
Mobile network subscribers constitute the primary stakeholders of the proposed system, as they are the direct users of the mobile application and the main source of data collected. These users interact with the application in both active and passive ways, making their role central to the success of the system.

From an active perspective, users provide subjective feedback on their network experience through the application interface. This includes rating service quality based on parameters such as:

- Call clarity and reliability
- Internet speed and responsiveness
- Frequency of call drops or connection failures
- Overall satisfaction with the network

Such feedback is critical because it reflects the Quality of Experience (QoE), which cannot always be accurately measured using technical metrics alone.

From a passive perspective, the application automatically collects objective data from users' devices without requiring continuous manual input. This includes:

- Signal strength
- Network type (e.g., 3G, 4G, 5G)
- Latency and bandwidth
- Location and timestamp

This dual data collection approach ensures that both user perception (subjective data) and actual network performance (objective data) are captured, providing a more comprehensive understanding of network quality.

Additionally, mobile network subscribers influence the system in several important ways:

- Their level of participation determines the volume and reliability of collected data

- Their feedback patterns help identify common issues and trends
- Their acceptance or reluctance affects the adoption and scalability of the application

To ensure effective user participation, the application must be designed to be:

- User-friendly, requiring minimal effort to interact with
- Non-intrusive, running efficiently in the background
- Secure, ensuring privacy and protection of personal data
- Beneficial, offering users insights or improvements based on their input

2) Secondary Stakeholders

Mobile Network Operators (MTN, Orange, Camtel)

Mobile network operators are key organizational stakeholders that directly benefit from the insights generated by the proposed system. They are responsible for providing communication services and maintaining network infrastructure, and thus have a strong interest in understanding and improving the Quality of Experience (QoE) of their subscribers.

The data collected by the mobile application serves as a valuable resource for operators in several ways. First, it provides real-time visibility into network performance, enabling operators to identify issues such as poor signal strength, congestion, and service outages. Second, by combining subjective user feedback with objective performance metrics, operators gain a comprehensive understanding of customer satisfaction, which is often difficult to measure using traditional monitoring tools alone.

Additionally, mobile network operators can use this data to:

- Optimize network resource allocation and bandwidth usage
- Detect and respond to faults more quickly
- Improve service reliability and coverage
- Enhance customer experience and reduce churn

Furthermore, the insights derived from user data can support strategic decision-making, such as infrastructure upgrades, deployment of new technologies, and targeted improvements in high-demand areas.

System Developers

System developers are the stakeholders responsible for the design, implementation, testing, and maintenance of the proposed mobile application. Their role is critical in translating user requirements and system specifications into a functional and reliable software solution.

During the development phase, developers are tasked with:

- Designing the system architecture
- Developing the mobile application interface

- Implementing background data collection mechanisms
- Ensuring seamless communication between the app and backend systems

They must also ensure that the application is:

- Efficient, with minimal impact on device performance and battery life
- Secure, protecting user data through encryption and safe storage practices
- Scalable, capable of handling a growing number of users
- User-friendly, providing an intuitive interface for easy interaction

In addition, developers are responsible for system testing and debugging, ensuring that the application functions correctly under various conditions. After deployment, they continue to maintain the system by:

- Fixing bugs
- Updating features
- Improving performance based on user feedback

Thus, system developers play a central role in ensuring that the application meets both technical requirements and user expectations, contributing to the overall success of the project.

Regulatory Authorities

Regulatory authorities are responsible for ensuring that the system operates within the legal and ethical framework governing telecommunications and data usage. In the context of this project, they play an important role in overseeing data privacy, security, and compliance.

Given that the application collects user-related data such as location, network usage, and performance metrics, it is essential to adhere to regulations that protect user rights and prevent misuse of information. Regulatory authorities ensure that:

- User data is collected with proper consent
- Data is stored and processed securely
- Privacy policies are clearly defined and enforced
- The system complies with national and international data protection laws

They may also establish guidelines regarding:

- Data sharing between operators and third parties
- Ethical use of collected information
- Transparency in how user data is utilized

Compliance with these regulations not only protects users but also enhances trust and credibility in the system, encouraging greater user participation.

III. REQUIREMENT GATHERING TECHNIQUES

To ensure that the system requirements are accurate, comprehensive, and aligned with user needs, multiple requirements gathering techniques were employed. Using a combination of methods helps to capture both quantitative data (numbers, trends) and qualitative insights (opinions, experiences), leading to a more robust system design.

1) Surveys

A structured survey was developed using Google Forms to systematically collect user experience data from mobile network subscribers. The survey was carefully designed to ensure clarity, ease of use, and relevance to the objectives of the project. It consisted mainly of close-ended questions to facilitate efficient data analysis, while also capturing key aspects of user interaction with mobile networks

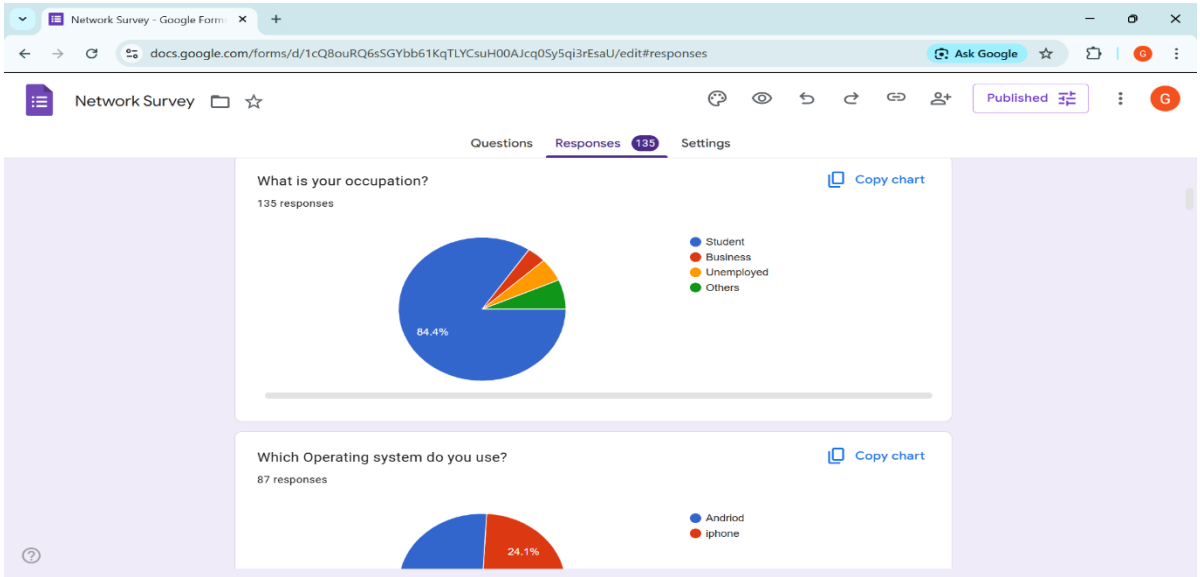
Purpose:

The survey served as the primary data collection tool for this study. Its main objectives were to:

- Gather quantitative data from a large and diverse group of users, enabling statistical analysis of network performance.
- Identify common issues and patterns experienced by subscribers across different networks.
- Understand user demographics and how they relate to network usage behavior.
- Provide a foundation for data-driven decision-making in the system design phase.

Data Collected:

- Age range
- Occupation
- Mobile network provider
- Operating system
- Network performance ratings
- Call drop experiences
- Location of network usage



2) Interviews

Interviews were conducted with selected mobile network subscribers to obtain in-depth and qualitative insights into their experiences with mobile network services. Unlike surveys, which provide structured and quantitative data, interviews allow for a more flexible and detailed exploration of user opinions, perceptions, and expectations.

Purpose of Interviews

The primary objectives of conducting interviews were to:

- Gain a deeper understanding of user frustrations and challenges with mobile networks.
- Explore user expectations for improved network performance.
- Identify issues that may not be captured through structured survey questions.
- Validate and complement the findings obtained from the survey.

Interview Process

A small group of users representing different demographics like students, professionals, and business users were selected for the interviews. The interviews were conducted in an informal and conversational manner to encourage participants to express their experiences freely.

Sample Questions Asked

The following questions were used to guide the interviews:

- What frustrates you most about mobile network services?
- Have you ever experienced frequent call drops or slow internet speed?
- Have you ever switched network providers due to poor service?
- Would you use a mobile application that monitors your network performance?
- What features would you expect from such an application?

The interviews revealed several important insights like:

- Many users experience inconsistent network performance, especially during peak hours
- Call drops and slow internet speed are common complaints
- Some users have switched network providers, indicating customer churn due to dissatisfaction
- Users expressed interest in a system that can:
    - Monitor network performance in real time
    - Provide feedback and recommendations
    - Help them make better network choices

3) Brainstorming

Brainstorming sessions were conducted as part of the requirement gathering process to generate innovative ideas and potential solutions for the proposed mobile application. This technique involves collaborative thinking, where we freely share ideas without immediate criticism, allowing for a wide range of possibilities to be explored.

Purpose of Brainstorming

The main objectives of the brainstorming sessions were to:

- Identify key features required in the mobile application.
- Explore possible solutions to network-related challenges.
- Encourage creative and critical thinking among participants.
- Develop ideas that align with both user needs and system goals.

Process of Brainstorming

The sessions involved group discussions where we (potential users and developers), contributed ideas based on our experiences with mobile networks. All ideas were initially recorded, after which they were evaluated and refined based on feasibility, relevance, and impact.

Key Ideas Generated

The brainstorming process led to the identification of several important features for the proposed system, including:

- Real-time network performance monitoring.
- Automatic data collection (signal strength, speed, latency).
- User feedback system for rating network quality.
- Alert and notification system for poor network conditions.
- Data visualization dashboard for easy interpretation of results.
- Location-based performance tracking.

4) Reverse Engineering

Reverse engineering was employed as a requirement gathering technique to analyze existing mobile network monitoring applications and systems. This approach involves studying already developed solutions in order to understand their structure, functionality, strengths, and weaknesses. By doing so, the project benefits from prior work while identifying areas where improvements can be made.

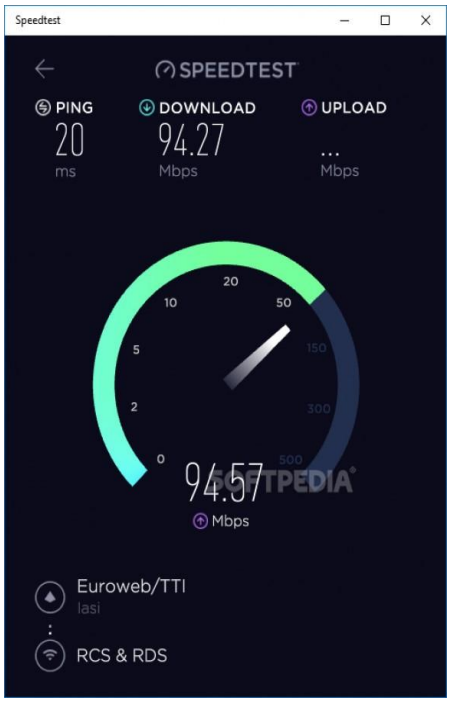
Purpose of Reverse Engineering

The main objectives of applying reverse engineering in this project were to:

- Avoid reinventing the wheel by leveraging existing technologies and solutions.
- Identify best practices used in current network monitoring applications.
- Understand the strengths and limitations of existing systems.
- Discover opportunities for innovation and improvement.
- Ensure that the proposed system is both competitive and relevant.

Systems Analyzed

Various existing mobile applications and tools related to network performance monitoring were reviewed such as **Speed Test, Opensignal, nperf**.



These are applications that measure:

- Signal strength
- Internet speed
- Latency and connectivity
- Network availability

Features Identified

Through analysis of these systems, several useful features were identified:

- Real-time display of signal strength and network type (3G/4G/5G)
- Internet speed testing functionality
- Simple and user-friendly interfaces
- Graphical representation of performance metrics
- Basic logging of network activity

IV. DATA GATHERING

Data gathering for this project was carried out primarily through the use of a structured survey distributed via an online platform (Google Forms). This approach enabled the efficient collection of data from a diverse group of mobile network subscribers, ensuring that the dataset was both representative and relevant to the objectives of the study.

Key Points

- A total of **133 responses** were successfully collected from participants
- The data captured included both:
  - **Subjective data**, which reflects users' personal experiences, opinions, and satisfaction levels regarding mobile network performance
  - **Objective data**, which represents measurable usage patterns such as device type, network provider, and usage context

1) Types of Data Collected

a) Subjective Data (User Feedback)

This type of data provides insight into how users perceive network performance. It includes:

- Network experience ratings
- Call drop occurrences
- Overall satisfaction with network services

Subjective data is important because it captures the Quality of Experience (QoE), which cannot always be measured through technical parameters alone.

b) Objective Data (Usage Patterns)

Objective data represents factual and measurable information about user behavior and device characteristics. It includes:

- Age group and occupation
- Operating system used (Android/iOS)
- Network provider (MTN, Orange, Camtel)
- Location of network usage

This type of data helps in identifying patterns and correlations that influence network performance and user experience.

V. DATA CLEANING

Before performing any form of analysis, the collected dataset was carefully cleaned to ensure accuracy, consistency, and reliability. Data cleaning is a crucial preprocessing step that improves the quality of the dataset and prevents misleading results during analysis.

Purpose of Data Cleaning

The primary objectives of data cleaning were to:

- Eliminate errors and inconsistencies in the dataset.

- Ensure uniformity in responses.
- Prepare the data for effective analysis and visualization.
- Improve the overall reliability of the results.

1) Steps Involved

The following steps were carried out during the data cleaning process:

a) Removal of Duplicate Entries

Duplicate responses were identified and removed to avoid over-representation of certain data points. This ensured that each response was counted only once, maintaining the integrity of the dataset.

b) Correction of Inconsistent Responses

Some responses contained inconsistencies in formatting, such as variations in text entries like mtn, MTN, Mtn. These were standardized into a uniform format to ensure accurate grouping and analysis.

c) Handling Missing Values

Missing or incomplete data entries were identified and addressed. Depending on their significance:

- Missing values were replaced with appropriate placeholders Unknown.

d) Formatting Data into Structured Tables

The cleaned data was organized into a structured tabular format using Microsoft Excel. This involved:

- Assigning clear and consistent column headers
- Converting the dataset into Excel tables
- Ensuring proper data types like text, numbers, dates.

This structure enabled efficient use of analytical tools such as Pivot Tables and charts.

2) Tools Used

The following tools were used during the data cleaning process:

- **Microsoft Excel:**
  Used for editing, formatting, and organizing the dataset
- **Pivot Tables:**
  Used to identify inconsistencies, detect duplicates, and summarize data for validation.

VI. USER RELUCTANCE ASSESSMENT

One of the key challenges in deploying a mobile application for collecting user experience data is user reluctance. Despite the potential benefits of such a system, users may be hesitant to adopt or continuously use the application due to various concerns. Understanding and addressing these concerns is essential to ensure high user participation, data reliability, and system success.

Several potential issues that may discourage users from engaging with the system were identified such as:

1) Privacy Issues

Users were concerned about how their personal data is collected, stored, and used. Since the application may access sensitive information such as location and network usage, users were afraid of:

- Unauthorized access to their data
- Misuse of personal information
- Lack of control over data sharing

2) Battery Consumption

Continuous background monitoring of network parameters may lead to increased battery usage. Users are often reluctant to install applications that:

- Run persistently in the background
- Reduce device battery life
- Affect overall device performance

3) Data Usage

The application may require periodic data transmission to send collected information to a central server. This raises concerns about:

- Increased mobile data consumption
- Additional costs for users with limited data plans

Mitigation Strategies

To overcome these challenges and encourage user adoption, the proposed system incorporates the following strategies:

1. Ensuring Data Privacy and Security

- Implement strong data encryption during transmission and storage.
- Provide clear privacy policies explaining how data is used.
- Allow users to control what data is shared.
- Ensure compliance with data protection regulations.

2. Minimizing Battery and Resource Usage

- Optimize the application to run efficiently in the background.
- Reduce the frequency of data collection when not necessary.
- Use energy-efficient algorithms to minimize battery drain.

3. Reducing Data Consumption

- Compress data before transmission.
- Limit the amount of data sent to essential information only.
- Allow users to restrict data transmission to Wi-Fi networks.

4. Providing Clear User Benefits

- Offer users insights into their network performance.
- Provide alerts for poor network conditions.
- Suggest better network options when necessary.
- Demonstrate how their data contributes to improving network services.

5. Maintaining a User-Friendly Interface

- Design a simple and intuitive application interface.
- Minimize user interaction requirements.
- Ensure smooth and seamless user experience.

VII. CONCLUSION OF REQUIREMENT GATHERING

The requirement gathering phase successfully identified and documented the needs, expectations, and challenges of both users and system stakeholders. By employing a combination of techniques including surveys, interviews, brainstorming, and reverse engineering the project was able to capture both quantitative data and qualitative insights, ensuring a comprehensive understanding of the problem domain.

This multi-method approach provided a balanced perspective by integrating user experiences (subjective data) with measurable usage patterns (objective data). As a result, the project established a strong and reliable foundation for subsequent stages of system development, particularly in analysis, design, and implementation.

Impact of Requirement Gathering

The insights obtained during this phase played a critical role in shaping the overall direction of the project. Specifically, they directly influenced:

a) Features of the Proposed Mobile Application

User feedback and identified challenges guided the selection of key system features, such as:

- Real-time network performance monitoring.
- Automatic data collection.
- User feedback and rating mechanisms.
- Alert and notification systems.

These features were designed to address the actual needs and expectations of users.

b) Structure of Data Analysis

The type of data collected determined how the dataset was organized and analyzed. It influenced:

- The use of structured tables and pivot tables
- The selection of appropriate charts for visualization
- The identification of meaningful patterns and trends

This ensured that the analysis was both relevant and insightful.

c) Overall System Architecture

The understanding of system requirements guided the design of the system architecture, including:

- Data collection mechanisms (mobile application)
- Data processing and storage components
- Communication between user devices and central system

This resulted in a design that is both efficient and scalable, capable of supporting real-time data collection and analysis.

In conclusion, the requirement gathering phase was instrumental in aligning the system design with real-world user needs and technical requirements. It ensured that the proposed solution is user-centered, data-driven, and practically applicable, thereby increasing the likelihood of successful implementation and adoption.